

UTILITIES

Polymath

Source: README.md

Knowledge, beautifully organized. A free, fast, open knowledge hub covering AI, machine learning, data science, mathematics, physics, cryptography, finance, science, technology, and history — with reactions, comments, and a community forum.

🔗 **Live site:** <https://ankitgupta1729.github.io>

Built by [Ankit Gupta](#). Content is licensed CC BY 4.0; code is MIT.

🌟 Features

- **Beautiful, responsive design** with light/dark mode and tasteful motion
- **Articles** in Markdown/MDX across 10 knowledge domains, with categories, tags, and reading time
- **Topic filtering & full-text search** (Pagefind — runs entirely in the browser, no server)
- **Custom comment system** — beautiful in-app threaded comments, replies, likes & sorting (Supabase-backed, with a local/demo fallback)
- **In-browser AI assistant (“Ask Poly”)** — runs free Hugging Face models on-device via Transformers.js; retrieves relevant articles and writes grounded answers. No API key, no cost.
- **Reactions** (👍 👎 ❤️), per-article like counter, bookmarks & reading list
- **Community page + forum** with member profiles, leaderboard and activity feed
- **Social sharing** (X, LinkedIn, Facebook, Reddit, HN, Telegram, WhatsApp, email, native share) + RSS feed
- **Auto-generated social preview (OG) images** per article + SEO (JSON-LD, sitemap, canonical URLs)
- **\$0 hosting** on GitHub Pages, deployed automatically via GitHub Actions

🔧 Tech stack

Concern	Choice
Framework	Astro 5 (static output)
Styling	Tailwind CSS 4 + Typography plugin
Content	Markdown / MDX content collections
Search	Pagefind
Comments / likes	Custom UI + Supabase (optional) / localStorage
AI assistant	Transformers.js (on-device, free)
Hosting	GitHub Pages (free)
CI/CD	GitHub Actions

🚀 Local development

BASH

```
npm install
npm run dev      # start dev server at http://localhost:4321
npm run build    # build to ./dist and generate the search index
npm run preview  # preview the production build locally
```

✍️ Adding an article

Create a Markdown file in `src/content/blog/`:

MARKDOWN

```
---
title: "Your Title"
description: "One or two sentence summary."
category: "ai"          # see slugs in src/consts.ts
tags: ["tag1", "tag2"]
pubDate: 2026-06-17
heroEmoji: "🤖"
featured: false         # set true to highlight on the home page
---

Your content here...
```

Then publish (see deployment below).

■ 📦 Deployment

The live site is published to the `gh-pages` branch and served by GitHub Pages. To rebuild and redeploy after changing content:

```
npm run deploy    # builds, generates the search index, and pushes ./dist to gh-pages
```

BASH

Optional: enable fully automated CI deploys

A ready-made GitHub Actions workflow lives at `.github/deploy.yml.disabled`. Pushing files under `.github/workflows/` requires a token with the `workflow` scope, so to activate it once:

```
gh auth refresh -h github.com -s workflow    # one-time, opens browser
git mv .github/deploy.yml.disabled .github/workflows/deploy.yml
git add .github && git commit -m "Enable CI deploy" && git push
```

BASH

Then set **Settings** → **Pages** → **Source** → **GitHub Actions**. After that, every push to `main` builds and deploys automatically — no manual step.

■ 💬 Comments (custom, no third party)

Every article and the forum use a **custom, in-app comment box** — threaded replies, likes, sorting and avatars. Out of the box it works immediately with sample discussion + your own comments saved in the browser. To make comments **shared and permanent across all visitors**, connect Supabase (free) — see the storage section below.

■ 🤖 AI assistant (“Ask Poly”)

A floating assistant on every page runs **free Hugging Face models entirely in the visitor’s browser** via [Transformers.js](#) — no API key, no server, no cost. It embeds the article index on-device (`all-MiniLM-L6-v2`), retrieves the most relevant pieces, and writes a grounded answer (`LaMini-Flan-T5-77M`), citing the source articles. The first question downloads the models once (~100 MB) and caches them; afterwards it works instantly and offline. Configure or disable it via `AI` in `src/consts.ts`.

■ 💾 Data & storage (how “saving things” works)

The site is **static**, so there’s no server to maintain — yet it still stores plenty:

Need	Where it's stored	Notes
Reactions, your own comments, likes, reading list, newsletter intent	Browser localStorage	Instant, private to each visitor, zero cost.
Shared, permanent comments & like counts	Supabase (optional, free tier)	Real database; off by default.
Demo community (members, sample threads)	<code>src/lib/community.ts</code>	Seed data, clearly labelled "Demo" in the UI. Edit freely.

Want shared comments + like counts across all visitors? Add [Supabase](#) (free tier). The comment box and ❤️ like button already support it — they use a per-visitor localStorage fallback until you add keys, then go global automatically.

1. Create a Supabase project. In the SQL editor, run:

```

-- Likes
create table article_likes (slug text primary key, likes int not null default 0);

create or replace function toggle_like(article_slug text, delta int)
returns int language plpgsql security definer as $$
declare new_count int;
begin
  insert into article_likes(slug, likes) values (article_slug, greatest(delta,0))
  on conflict (slug) do update set likes = greatest(article_likes.likes + delta, 0)
  returning likes into new_count;
  return new_count;
end; $$;

-- Comments
create table comments (
  id uuid primary key default gen_random_uuid(),
  slug text not null,
  parent_id uuid references comments(id) on delete cascade,
  name text not null,
  body text not null,
  likes int not null default 0,
  created_at timestampz not null default now()
);
alter table comments enable row level security;
create policy "read" on comments for select using (true);
create policy "insert" on comments for insert with check (char_length(body) between 1
and 1000);

create or replace function like_comment(comment_id uuid)
returns void language sql security definer as $$
  update comments set likes = likes + 1 where id = comment_id;
$$;

```

2. Paste your project URL + anon key into `SUPABASE` in `src/consts.ts` and redeploy.

The anon key is safe to ship in a static site; RLS keeps data locked down. This is **optional** — localStorage + the demo seed already make comments and likes work out of the box.

■ Custom domain (optional)

GitHub Pages supports a custom domain for free (you only pay your registrar for the domain itself, ~\$10/yr). Add a `CNAME` file in `public/` with your domain and configure DNS per [GitHub's guide](#).

■ License

- **Code:** MIT
- **Content:** [CC BY 4.0](#)